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Power cable vibration monitoring based on wireless distributed sensor network

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Abstract

Mechanical damage caused by construction, theft and transformation, has caused a long-term threat to the safe operation of cables. Owing to the complexity of cable distribution and operating environment, it seems impossible to apply traditional online monitoring systems with active alarms to power cable systems, especially considering the high cost of conversion and maintenance. In this paper, an integrated smart sensors based on MEMS accelerometers is proposed with the advantages of the self-organizing network, low latency, low power consumption according to the concept of Internet of Things, and a distributed cable monitoring method is described. Then, a diagnosis method based on the time-frequency characteristic parameters is exemplified.

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Keywords: Wireless sensing network; cable damage monitoring; Internet of Things, intelligent sensor

1. Introduction

Power cables have gradually replaced overhead lines due to their advantages of less occupied area and good environment immunity [1]. In recent years, the ever-increasing power grid cable usage makes the resources of underground cable channels become more and more strained. In this case, the problems of fire prevention and

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external damage are becoming increasingly prominent [2]. Besides, the external damage of cables due to construction, theft and reconstruction has long threatened the safety of cable operations. However, due to the widespread distribution of cables and the complex operating environment, there is currently no effective monitoring and warning method [3].

The traditional online monitoring system [4-6] usually includes signal coupling unit, analog pre-processing unit, power supply unit, transmission unit, data acquisition and remote system (e.g. SCADA) and various data interfaces. However, the above monitoring system framework does not meet the requirements of cable monitoring, owing to the complex feed network, long-span distribution, and huge cost. In addition, the numerous monitoring access nodes pose huge challenges for data storage, information matching, and data applications. Although distributed optical fiber vibration sensing technology based on OTDR (Optical Time Domain Reflectometer Technology) [7,8] has been applied, it is difficult to distinguish the type of external disturbance, and the system has a high false alarm rate and high costs of installation and post-maintenance. Therefore, in order to meet the development requirements of the smart distribution network, it is necessary to explore an efficient, reliable, and economical cable distributed antibreakage monitoring technology.

With the development of microelectromechanical (MEMS) sensor technology, the integrated design of system on a chip (SoC) [9] and edge computing [10] become possible, which can be used as a frontend sensor to monitor cable vibration sound waves caused by external force. In addition, the development and application of low-power, long-distance IoT communication technology (LPWA) is not only widely used in the consumer market, but also increasingly used in the industrial field. Due to its convenient and low-cost information network framework, it is extremely suitable for cable monitoring with large access volume, wide distribution area and cost-sensitive cases [1]. Based on IoT, this paper proposes a distributed cable acoustic wave vibration monitoring method. In order to meet the requirements of low-power and scalable monitoring networks, this paper first analyzes the sensor network framework and data flow method, and then discusses the key technical links of the distributed smart sensors based on the piezoelectric principle, including signal conditioning and power consumption management, data format and wireless communication; finally, based on the extraction of time-frequency characteristic parameters of cable vibration, a method for analyzing and warning the cable external force disturbance mode based on reduced state transmission volume is proposed, forming a flexible and scalable distributed based on the IoT Intelligent early-warning system for preventing cable damage.

2. Cable vibration sensing network framework

2.1. Overall framework of IoT-based sensor network

Considering that the distributed monitoring system should have good scalability and accessibility, this article will design the sensor network into four layers according to the characteristics of cable monitoring, namely the sensor layer, network layer, data layer and application layer. As shown in Fig. 1.

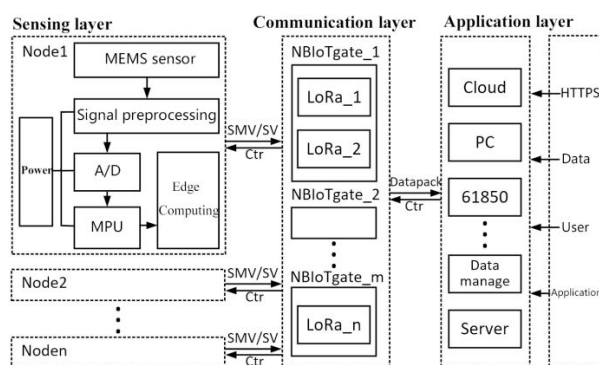


Fig. 1. Overall framework of sensing network.

The sensor layer is the physical foundation of the Internet of Things for cable vibration status monitoring. Its main functions are: 1) Monitoring the external force disturbance status of the cable and sending the information to the communication network; 2) Assigning the host computer to execute the command (Ctrl) to the sensor Node; 3) Calculate the collected data (SMV) based on the frontend MPU to form a reduced state quantity (SV).

The communication layer is the signal channel of the data up/down link, which is composed of a low power consumption, wide area network based IoT network transceiver [11]. The transceiver unit transfers the SMV, SV, and Ctrl information obtained by the sensing layer to the "two-way node-server" or "node-gateway-server" mode for millisecond-level information transmission

The application layer is used to implement protocol transmission (HTTPS), computer access (PC), server access (Server), data management (Data manage), and provides scalable with standard communication protocols. At the same time, this layer responds to data access requirements via the mapping interface (IEC61850) and can also establish a scalable database in the central server.

2.2. Data communication method based on LPWA

Taking into account the complexity of distributed monitoring, installation objects, operating environment, and electromagnetic interference, a reliable data communication method is critical to the operational efficiency and stability of the sensor network. Data transmission should be designed to fulfill the specific requirements of sensor layer data interaction, such as data transmission rate, channel bandwidth, delay, flow logic, and coverage. For example, the sensors are usually installed along a cable system which is distributed in the area of several kilometers to tens of kilometers. In this case, low-power wide area network (LPWAN) technology becomes the preferred solution. By fully considering the local stability, wide-area stability, power consumption, and economy of the local discharge monitoring of cable accessories, in this paper, LoRa and NB-IoT are selected as the primary choices for forming a sensor network.

Due to the decentralized deployment of cable sensor nodes, it is difficult to carry out information exchange through point-to-station mode. In this paper, LoRa nodes are used as the communication base layer, and NB-IoT base stations are used as data and wide area interactive nodes to achieve LoRa communication with sensor nodes and NB-IoT communication with a base station (or cloud) [12]. On this basis, not only can the sensor node information be collected, but also the conversion between LoRa and NB-IoT can be easily achieved. In order to reduce the channel transmission pressure, central calculation pressure and sensor power consumption, the sensor layer needs to maintain a low frequency of data interaction, so the up-stream data transmission of the sensor node only occurs when the warning threshold is exceeded, and close the upstream channel after receiving the downlink control command or set the waiting time to close automatically, and the sensor setting command is only issued within the communication effective period, which not only ensures the realtime status warning, but also can minimize the maximum communication pressure and redundant data transmission. Fig.2 shows the data transfer method and communication method between the sensor LoRa node and the LoRa-NB-IoT gateway.

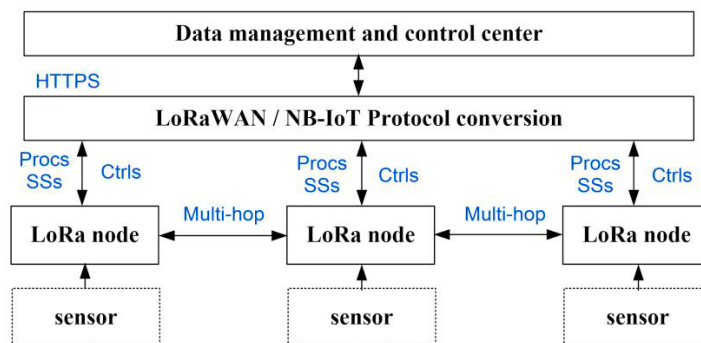


Fig. 2. Data transfer and conversion in WSN.

3. Smart sensor node design

3.1. Signal coupling module

When the cable is damaged by external force, there are several typical types of signal sources, including air pick signals, mechanical excavation or pile driving signals. These external force signals are distributed in the frequency range of hundreds to kilohertz. Therefore, in this paper a MEMS-based acceleration sensor (ADXL1002) is developed as the acoustic wave coupling front-end, which has a wide linear frequency response (DC-11 kHz@3dB), large measurement range ($\pm 50g$), small temperature coefficient (5%, $-40^{\circ}\text{C}\sim 125^{\circ}\text{C}$), low typical power consumption (1.0 mA), low thermal noise density ($30 \mu\text{g}/\sqrt{\text{Hz}}$) and small size ($<5 \text{ mm}^3$).

The built-in amplifier module of the sensor has a primary sensitivity of 40mV/g, and the amplification gain is fixed at 0dB, and the output voltage signal amplitude is linearly related to the vibration acceleration. The relationship between the output voltage U , amplification gain A and the vibration acceleration a is expressed by

$$a = \frac{U}{s \times A} \quad (1)$$

3.2. Signal processing module

In order to improve the sensitivity of the sensor in the actual cable vibration measurement, the charge amplifier circuit is integrated in the front-end circuit design to amplify the amount of charge generated by the weak vibration. After the vibration acceleration signal is obtained, the velocity signal is obtained through integral calculation as

$$v(t) = \int_{-\infty}^0 a(t)dt + \int_0^t a(t)dt = v_0 + \int_0^t a(t)dt \quad (2)$$

The main frequency of the vibration signal is in the range of tens of kHz, and the sampling and restoration of the signal can be achieved without a very high sampling rate. The amplification circuit of the original vibration signal is shown in Fig. 3, which uses a two-stage amplification circuit composed of LM324, the gain is set to 20dB, in order to eliminate DC offset, the cascade reduces the self-excitation through DC blocking processing. The analog bandwidth of the amplification unit is verified as 380kHz with satisfied ratio of signal to noise.

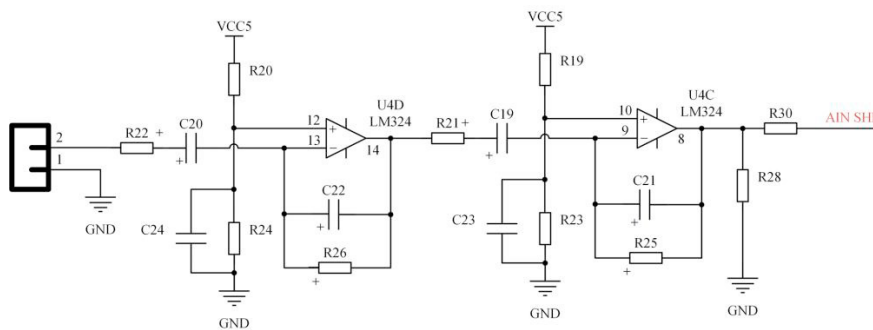


Fig. 3. Amplifying circuit of vibration signal.

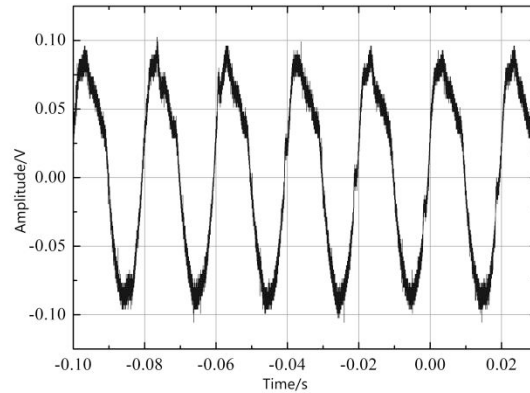


Fig. 4. Periodic vibration signal waveform measured by signal coupling unit.

It can be seen from the Fig.4 that the output signal amplitude of the piezoelectric sensing unit is stable and the frequency response is accurate. The 50Hz fundamental frequency signal contains the vibration signal of the cable itself and the harmonic components. After being calibrated by a standard vibration table, the sensor sensitivity $S = 25\text{mV/g}$, nonlinear error $\delta < 6.5\%$.

3.3. Sensor power management

This paper uses a current transformer (CT) to draw energy from the cable body to power the sensor energy storage unit. Fig.5 shows a schematic diagram of power capture based on CT.

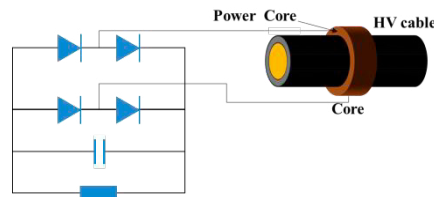


Fig. 5. Power acquiring from high voltage cable.

Let $N_1 (=1)$ and N_2 be the primary and secondary turns, $i_1(t)$ is the sinusoidal current on the high voltage side; $i_2(t)$ is the actual current flowing to the load; $i_m(t)$ is the excitation current, according to the electromagnetic induction law and the balance equation of the magnetomotive force of the transformer can be obtained as follows:

$$N_1 \cdot i_1(t) = N_2 \cdot i_2(t) + N_2 \cdot i_m(t) \quad (3)$$

wherein $i_1(t)/N_2$ is the equivalent current converted from primary to secondary, R_1 and L_1 are primary leakage resistance and leakage inductance, R_m and L_m are excitation resistance and inductance; R_2 and L_2 are secondary leakage resistance and leakage inductance; $U(t)$ is the voltage across the coil; C is the voltage stabilizing capacitor and u_0 is the load voltage. Due to the nature of the current source of the high-voltage cable, the primary leakage impedance is negligible; For the power coil, the secondary leakage impedance is very small relative to the excitation impedance, and thus can be ignored; considering the voltage drop of the rectifier bridge, the actual measured power data can be used in the actual power calculation plus the voltage drop to approximate the real output power, so it can also be ignored in the model; L_m is much larger than R_m in the excitation impedance, that is, the analysis of the output power of the power coil is mainly determined by L_m . In summary, the CT load working model can be simplified as shown in Fig.6.

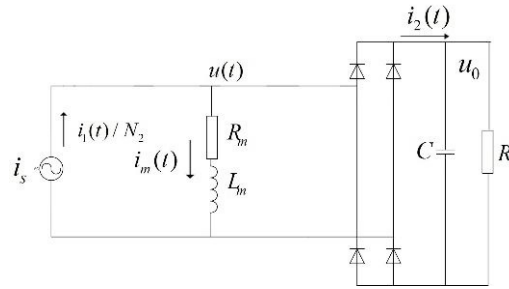


Fig. 6. Simplified circuit model with consideration of load.

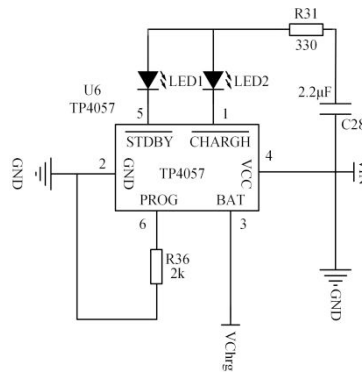


Fig. 7. Power management and voltage level control circuit

The piezoelectric sensor unit needs a stable constant current source for power supply. Since the sensor needs to work in the monitoring mode for a long time, the stability and continuity of the power supply is important, and the current carrying capacity of the cable directly affects the power input of the CT coil to draw power. Therefore, in order to avoid the load power supply interruption caused by the reduction of the current carrying capacity (working current is about 150A) or power failure, a lithium battery is used as a bypass backup power supply. In the signal preprocessing unit, a power management circuit is used to stabilize the output level, as shown in Fig. 7. The power management circuit can control the charge and discharge of the battery to prevent the battery from being overcharged and discharged.

In order to ensure that the output voltage and power meet the needs of the post-stage sensors, it is also necessary to design the corresponding power supply voltage regulation and start-up circuit, as shown in Fig.8. The control loop adopts the hysteresis comparison method, which improves the control speed and reduces the output voltage ripple. The output voltage of the first-stage pre-regulation circuit is 24V, which is reduced to 5V through the second-stage circuit for the load. The self-starting circuit can take the two-stage output voltage 24V and 5V as input and output the power supply voltage required by the control loop chip. At the same time, the voltage level monitoring is used to start the power output of the lithium battery to the load during the voltage loss start-up, and the self-start circuit is controlled to charge the lithium battery via the Buck regulating circuit.

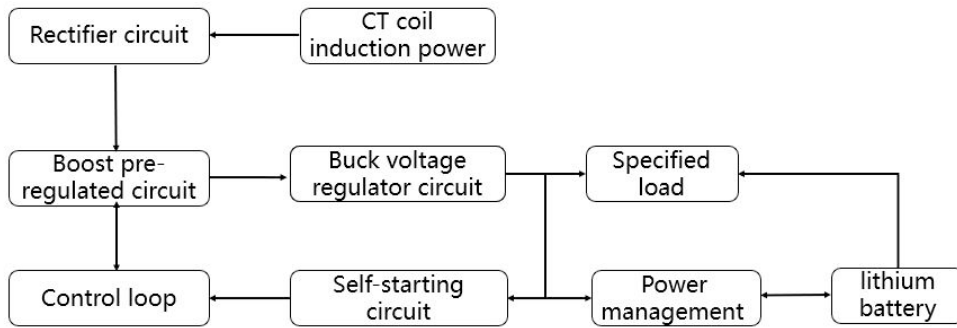


Fig. 8. Operation principle of voltage stabilization and start-up circuit

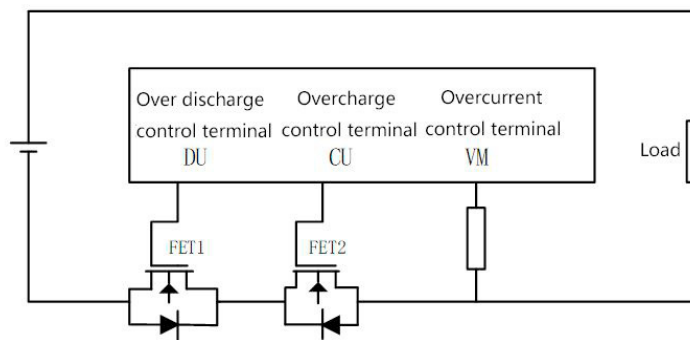


Fig. 9. Battery charge and discharge protection circuit

According to the energy loss of each module, the total power consumption is calculated for battery life of 48 hours. Because repeated charge and discharge of the battery will cause capacity degradation due to the chemical side reactions inside the battery, a protection circuit is designed to prevent serious battery loss. Two external MOS tubes (FET1, FET2) are integrated on another chip as a switch. When the discharge current is too large, FET1 for discharge control is turned off, and the battery is prohibited from discharging to the load. When the battery is charged to the over-charge threshold voltage, FET2 is turned off, prohibiting the current flowing into the battery. When the battery voltage drops to the overdischarge threshold voltage, the discharge control FET1 is turned off, prohibiting the battery from discharging to the load. Fig.9 shows the battery charge and discharge protection circuit.

4. Simplified status data transmission

4.1. Uplink/downlink transmission

LoRa modems use proprietary modulation and demodulation procedures to combine spread-spectrum modulation and cyclic error correction coding techniques, thereby increasing link budget and anti-interference. LoRa modulation and demodulation technology can be optimized by adjusting the spreading factor (SF value) and modulation bandwidth (BW). In order to ensure a lower data transmission size and reduce the pressure of the communication unit and centralized calculation, the data transmitted to the gateway through the sensor is the simplified status data. At the same time, in order to meet the requirements of real-time and power consumption, a breakpoint monitoring strategy is applied to practical use. The monitoring density of the sensor N/s is determined by the modulation bandwidth and spreading factor, shown as [10].

$$N = T^{-1} \cdot \frac{SF}{2^{SF} \cdot BW^{-1}} \quad (4)$$

where T is the detection period of the analog signal of the sensor, which is 25 μ s in this article; SF is the spreading factor, 10 is the typical value in this article; BW is the modulation bandwidth, 125 kHz.

4.2. Simplified status data of vibration signal

Different from the analysis of power frequency voltage signal, the correlation between vibration signal and power frequency excitation is weak, and the vibration excitation is mostly caused by external force disturbance. The randomness of the signal source and the propagation path is strong, and the phase statistical information is difficult to apply. Therefore, this paper uses the quadratic short-time Fourier transform as a kernel function, and performs a time-frequency joint analysis on the vibration transmission signal to find time-frequency characteristic parameters.

If a window function with a short time width is given and the window slides, the quadratic short-time Fourier transform (STFT) $S_z(t, f)$ of the signal $z(t)$ is shown as

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Bulleted lists may be included and should look like this:

$$S_z(t, f) = |STFT_z(t, f)|^2 = \left| \int_{-\infty}^{\infty} z(t') \eta^*(t' - t) e^{-j2\pi f t'} dt' \right|^2 \quad (5)$$

$S_z(t, f)$ is the square of the short-time Fourier transform STFT(t, f) mode, which reflects the energy distribution of the signal in the time-frequency plane. Due to the existence of the window function, the quadratic short-time Fourier transform has local characteristics, which are both a function of time and a function of frequency. For a given time t , $S_z(t, f)$ can be regarded as the frequency spectrum at that moment. The quadratic STFT of the discrete time signal $z(n)$ can be shown as

$$S_z(n, \omega) = \left| \sum_{m=-\infty}^{\infty} z(m) \eta(n - m) e^{-i\omega m} \right|^2 \quad (6)$$

The discrete quadratic STFT sampling points at equal frequency intervals can be obtained from

$$S_z(n, r) = \left| \sum_{m=-\infty}^{\infty} z(m) \eta(n - m) e^{-i\frac{2\pi}{N} r m} \right|^2 \quad (7)$$

Let $m = l + n, l = k + sN, k = 0, 1, \dots, N - 1$, then equation (7) can be transformed into

$$S_z(n, r) = \left| e^{-i\frac{2\pi}{N} r n} \sum_{l=-\infty}^{\infty} z(l + n) \eta(-l) e^{-i\frac{2\pi}{N} r l} \right|^2 = \left| e^{-i\frac{2\pi}{N} r n} \sum_{k=0}^{N-1} z(k, n) e^{-i\frac{2\pi}{N} r k} \right|^2 \quad (8)$$

Through the above calculation, multiple time-frequency energy centers of vi-bration non-stationary signals can be obtained, which can be used as one of the characteristic parameters for judging the vibration mode.

5. Experiment results

During the experiment, the sensor is fixed on the cable body in the middle of the two brackets (as shown in Fig.10), and the simplified status data set output by the sensor is received through the gateway, and compared with the signal characteris-tic results obtained by the oscilloscope.

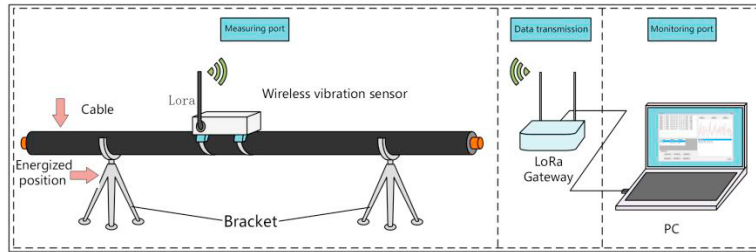


Fig. 10. Schematic diagram of test

Fig. 11 is the signal waveform of the electric drill when the impact is applied on the cable and the bracket.

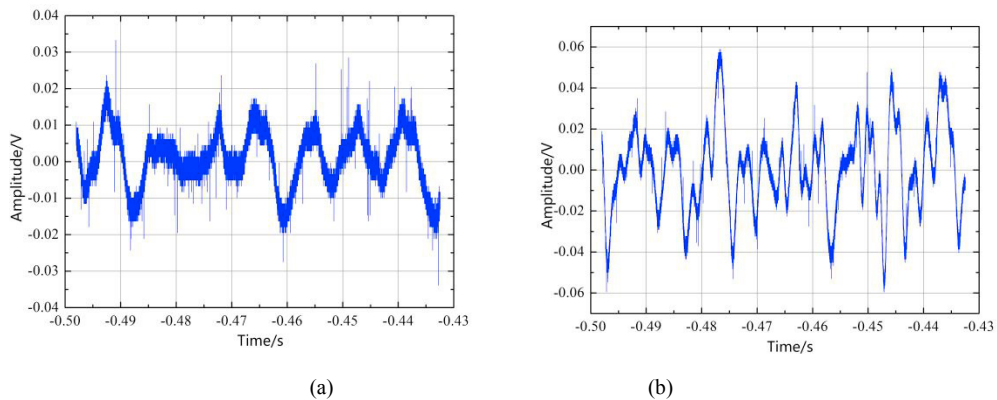


Fig. 11. Waveforms measured at different positions of the electrical drill. (a) the impact is applied on the cable; (b) the impact is applied on the bracket

It can be seen from the measured signals that the vibration signal on the ca-ble differs significantly under different construction conditions; when the vibration signal amplitude exceeds 0.1V, it can be considered that an external force damage event has occurred, and the sensor warning threshold can be set to 20dB;

Fig. 13 is the analysis result of the time-frequency characteristics of the electric drill excitation transmission. It can be seen from the figure that when the vibration of the electric drill is transmitted on the cable body, the low-frequency main component of the vibration amplitude shows irregular changes on the time axis, the sub-high frequency component is relatively stable on the time axis, and the higher frequency sub-component is relatively weak when the vibration of the electric drill is transmitted on the cable support, the low-frequency principal component of the vibration ampli-tude shows regular changes on the time axis, and the characteristics are similar to the former in the sub-high frequency and high-frequency regions.

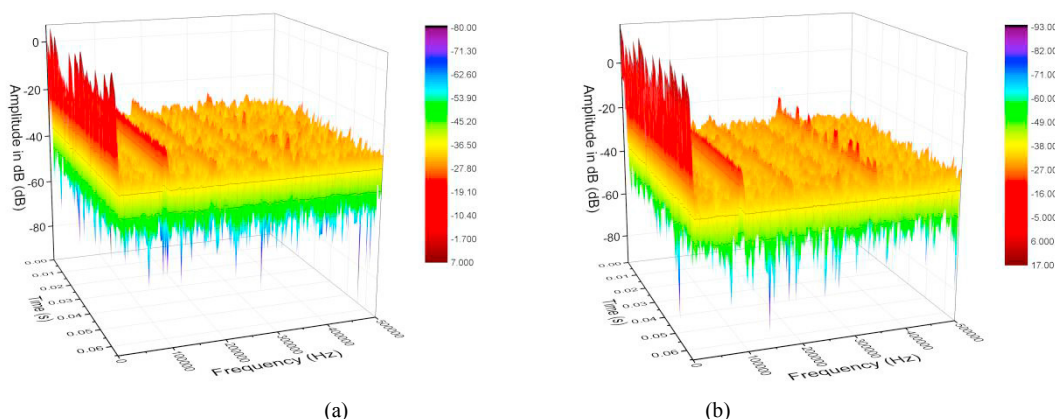


Fig. 12. Time-frequency analysis of electrical drill excitation. (a) the impact is applied on the cable; (b) the impact is applied on the bracket

Table 1 takes the electric drill applied to the cable as an example, and compares the characteristic quantity obtained by actual measurement data with the process quantity and error output by the sensor. It can be seen from the comparison that the errors between the process quantities used for sensor early warning and the actual measured feature quantities are within 10%, which can meet the external force monitoring and early warning during actual cable operation.

Table 1. Comparisons between actual Procs and output Procs (as electrical drill applied)

Process parameter	Actual value	Output value	Relative error (%)
Proc_acc	13.33	12.90	3.22
Proc_mag	25.51	23.12	9.37
Proc_freq	1	1	0
Proc_Eint	314.25	341.55	8.69
Proc_tmfm	[525.1,13.2]	[520.7,12.6]	[0.8,4.54]
Proc_t80	833	846	1.54
Proc_Rp	0.121	0.133	9.02

6. Conclusion

This paper first analyzes the sensor network architecture and data transfer method, and then discusses the key technical links of the distributed smart sensor based on the piezoelectric principle, including signal conditioning, power management, data format, and wireless communication; finally, according to the cable vibration Time-frequency characteristic parameter extraction, a method for analyzing and warning of cable external force disturbance mode based on reduced state transmission.

It can be seen from the comparison that the errors between the calculated process quantities used for sensor early warning and the actual measured feature quantities are within 10%, which can meet the external force monitoring and early warning during actual cable operation. The Internet of Things monitoring method proposed in this paper can provide new ideas for cable anti-breakage monitoring, and is of great significance for reducing the cost of cable operation and maintenance.

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